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Introduction

One of the most straightforward techniques for modulating a sinusoidal carrier wave is amplitude modulation (AM). The early 20th century saw the first usage of AM for radio speech signal transmission. However, AM is still utilized today in a variety of communication applications, such as two-way radios, broadcast radio, and aircraft VHF radios. This lab experiment will create AM signals, examine their characteristics in the time and frequency domains, and calculate their modulation indices. Students will also research how to demodulate AM signals using envelope and coherent detectors.

Procedure

The procedure of this experiment utilized the *Exp2_Setup.m* simulation MATLAB file.

Part A: Line spectra of periodic signals

- The goal is to see how waveform and duty cycle affect frequency components.
 1. Attach a spectrum analyzer to a signal generator.
 2. Square Wave Analysis: Configure the generator to produce a Square wave at 1kHz.
 3. Look at the spectrum; at 1kHz, there should be a single spike (line).
 4. Look for odd harmonics (3f, 5f, 7f, ...).
 5. Note the first three harmonics' power level (in dB).
 6. Duty cycle effect: Transform the signal into a pulse train.
 7. Change the duty cycle (e.g., 20%, 50%).
 8. When the duty cycle is an integer fraction, note how the line spectra's "sinc" envelope changes and how some harmonics vanish.

Part B: Modulation Index

- To see a continuous spectrum, this section switches from periodic signals to a single non-periodic event.
 1. Set the Signal Generator to generate a single rectangular pulse (or a pulse with a very long duration relative to the pulse width) to configure the pulse generator.
 2. Change the Pulse Width (τ) (Start with a broad pulse width)
 3. Examine the Spectrum Analyzer.
 4. Reduce the pulse width (τ) gradually.
 5. Make sure the spectrum expands in the frequency domain as the pulse narrows in the time domain (Examining the inverse relationship)
 6. Find the first null in the spectrum and confirm that it happens at $f = \frac{1}{\tau}$.
 7. Note the magnitude spectrum's continuous Sinc shape $\frac{\sin x}{x}$

Part C: Demodulation of AM Signals

- This section compares detection methods using the Exp2_EnvelopeAndCoherent.slx model Envelope detection (Hilbert vs. Rectifier))
 1. Launch the corresponding models of envelope detectors on Simulink.
 2. Enter a message signal $m(t)$ that falls inside the 2kHz bandwidth.
 3. Execute the Hilbert-based model and the Rectifier-based model.
 4. Examine the demodulated signal's smoothness and ripple using the scope.
 5. Keep in mind that additional filtering is needed to eliminate carrier ripple in the rectifier.

Theory

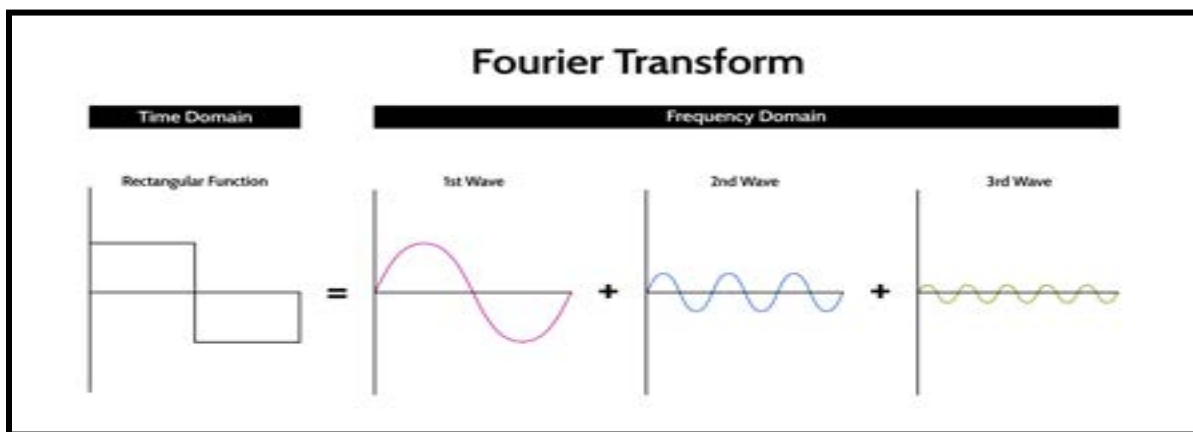


Figure 1: The Fourier Transform

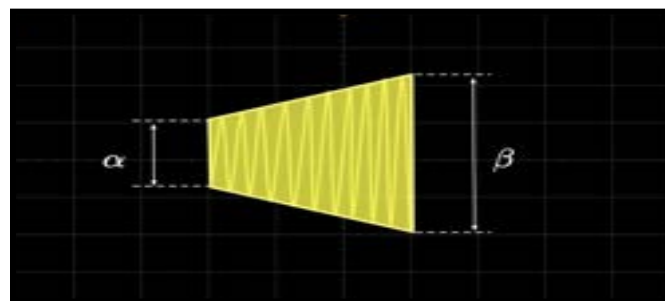


Figure 2: Trapezoidal Display used to Measure the Modulation Index μ

From the measurements we calculate:

$$A_{\min} = \frac{\alpha}{2}, \quad A_{\max} = \frac{\beta}{2}, \quad A_c = \frac{\beta + \alpha}{4}. \quad (\text{A.14})$$

Using the definition given in Equation (A.6) we determine the modulation index as:

$$\mu = \frac{\beta - \alpha}{\beta + \alpha}. \quad (\text{A.15})$$

Figure 3: The Modulation Index

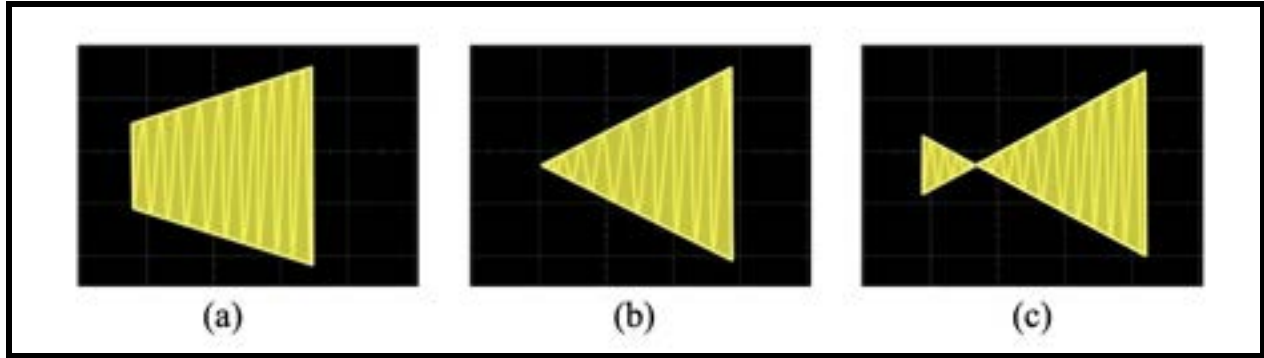


Figure 4: Trapezoidal Patterns for $\mu < 1$, $\mu = 1$, and $\mu > 1$ (Overmodulation)

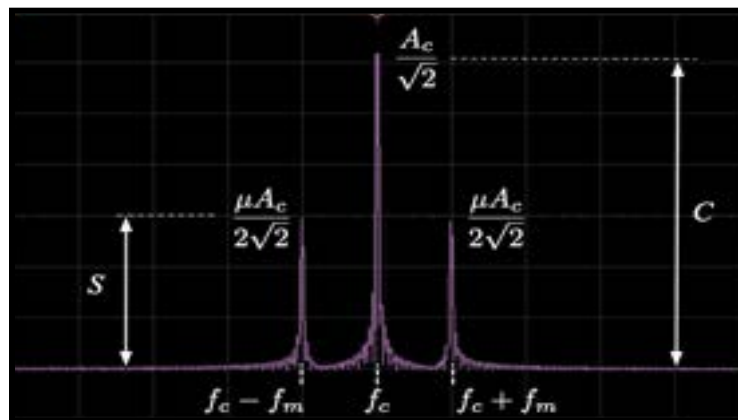


Figure 5: One-Sided RMS Spectrum of the Single-Tone Modulated AM Signal

$$A_c = \frac{1}{2}(A_{\max} + A_{\min}) \quad (\text{A.13})$$

$$\mu = 2 \frac{S}{C}, \quad (\text{A.16})$$

Figure 5: Ratio for the Modulation Index (μ)

List of Instruments

For this experiment, the following list of hardware equipment and software was utilized:

- Agilent DSO-X 2002A digital storage oscilloscope with waveform generation and spectrum analyzer options.
- GW Instek GFG-8216A function generator.
- Hewlett-Packard 33120A function/arbitrary waveform generator.
- A computer with a Linux operating system.
- Matlab/Simulink 2014b

PART A - Characteristics of AM Signals

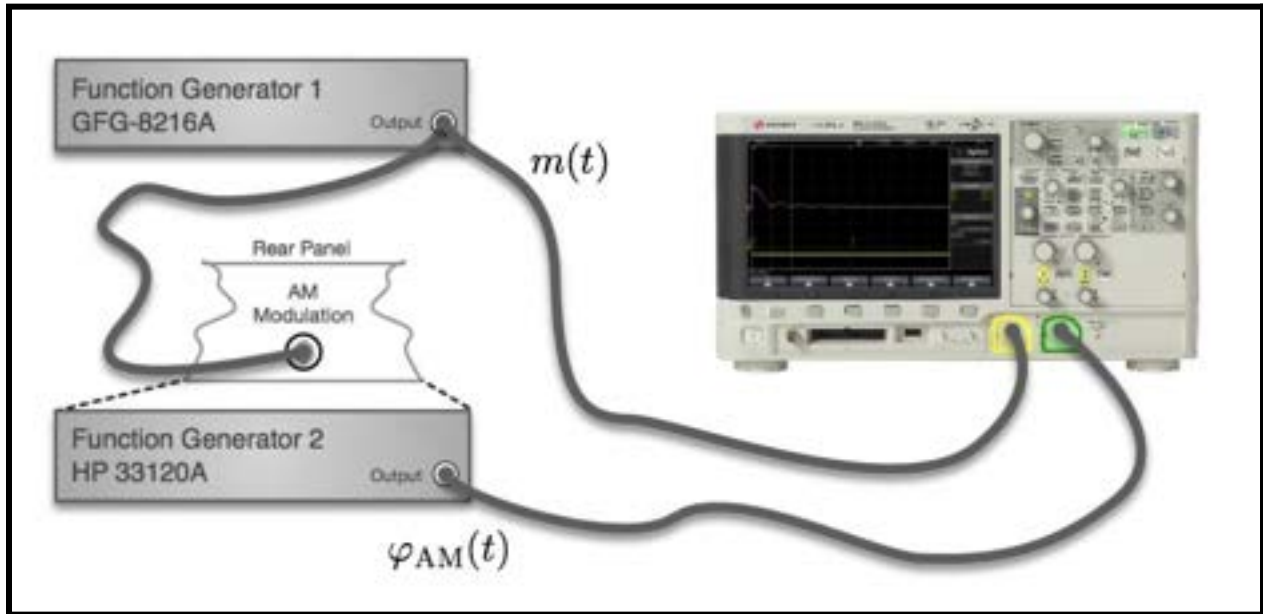


Figure A1.1: AM Signal Generation & Connection Diagram

$$\varphi_{AM}(t) = [A_c + A_m \cos(2\pi f_m t)] \cos(2\pi f_c t) \quad [1]$$

Equation A1.1: AM Signal with $A_m = 0.02A'_m$

PART B - Modulation Index

Problem B.1

Table B1.1 - Method 1

Amplitude V_{pp} (V)	A_{max} (V)	A_{min} (V)	A_c (V)	μ
4	2.04	-145 mV	0.9475	1.153034301
5	2.4	-155 mV	1.1225	1.138084633
6	3.10	-167mV	1.4665	1.113876577
8	3.9	-185 mV	1.8575	1.099596231
10	4.9	-210 mV	2.345	1.089552239

12	6.2	-231 mV	2.9845	1.077399899
14	7.4	-241 mV	3.5795	1.067327839
16	8.0	-241 mV	3.8795	1.062121407

Table B1.2 - Method 2

Amplitude V_{PP} (V)	Alpha (V)	Beta (V)	μ
4	0.125	0.3	0.4117647059
5	0.11	0.31	0.4761904762
6	0.08	0.33	0.6097560976
8	0.035	0.37	0.8271604938
10	0.005	0.425	0.976744186
12	0.035	0.45	0.8556701031
14	0.05	0.48	0.8113207547
16	0.05	0.47	0.8076923077

Table B1.3 - Method 3

Amplitude V_{PP} (V)	S (Vrms)	C (Vrms)	S (dB)	C (dB)	μ (for Vrms)	μ (for dB)
4	0.018	0.077	12.685	23.907	0.4675324675	0.5494523
5	0.02	0.077	8.12	16.32	0.5194805195	0.77809029
6	0.028	0.078	23.5	22.7	0.7179487179	2.192956392
8	0.034	0.078	15.625	23.125	0.8717948718	0.8433930069
10	0.043	0.081	17.5	21.8	1.061728395	1.219073794
12	0.05	0.078	18.7	23	1.282051282	1.219073794
14	0.028	0.112	15	26.7	0.5	0.5200319126
16	0.034	0.131	18.7	30	0.5190839695	0.5445402616

Based on the tabulated results, the modulation index μ for method 3 appears to be the most accurate. For method 1, for an increasing amplitude of V_{pp} , $\mu > 1$ in each case. This implies that the message signal is always overmodulated. Even before calculating μ this is noticeable because the collected values for A_{min} are below the zero crossing, which implies the phase of the carrier flips 180 degrees (phase reversal). Assuming that the data collected for Table B1.1 is accurate, then method 1 is unreliable for determining μ since it is not clear at which point the signal is overmodulated.

However, assuming that A_{min} was measured inaccurately and the values are actually non-negative, then it is expected that as V_{pp} increases, the signal increasingly becomes overmodulated. For method 2, as V_{pp} increases, μ increases towards 1. This is more accurate than the results in Table B1.2 because it is very clear that when the amplitude is increased, the signal becomes overmodulated.

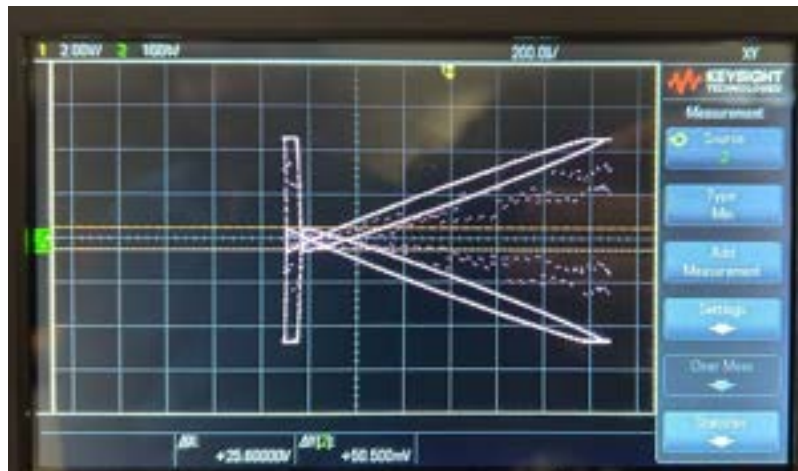


Figure B1.1 - XY mode, $V_{pp} = 14 \text{ V}$

At $V_{pp} = 14 \text{ V}$, on the oscilloscope, it is also clear that over-modulation eventually occurs because of the trapezoidal pattern. Therefore, method 2 is more accurate than method 1 because the modulation index increases towards 1.

The data collected in method 3, μ calculated from the V_{rms} values, show that as the amplitude increases, μ it becomes overmodulated. However, in comparison to the past two methods, when $V_{pp} = 10 \text{ V}$ the modulation index is greater than 1, it suddenly decreases despite the expectation that the signal is still overmodulated.

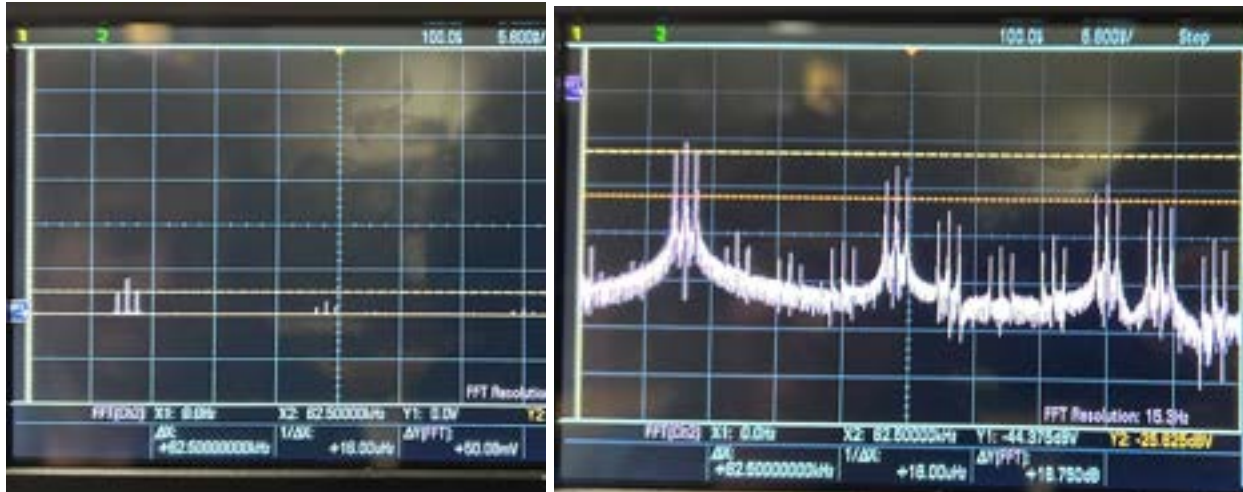


Figure B1.2 - $V_{pp} = 12\text{ V}$ for method 3 (Vrms & dB)

We notice that as the amplitude increases, the spectrum in dB becomes difficult to measure because of the significant distortion which creates sharp edges at various frequencies. This distortion also impacts the calculated index when measuring the carrier and sideband in Vrms.

Problem B.2

If the modulating signal $m(t)$ is arbitrary, then the most effective method for measuring μ is by using method 3. This method is the only one which utilizes the frequency domain to measure the sideband and carrier in dB. When measuring in dB, this method is capable of accurately measuring low modulation indices in comparison to time domain methods. Furthermore, when the signal is overmodulated, it is easier to notice the increase in sharp pulses which indicate that the signal is overmodulated.

Problem B.3

Table B3.1 - Power and efficiency calculations

Amplitude V_{pp} (V)	P_{SB} (W)	P_C (W)	Efficiency η (%)	μ (for Vrms)
4	0.000648	0.005929	9.852516345	0.4675324675
5	0.0008	0.005929	11.88883935	0.5194805195
6	0.001568	0.006084	20.4913748	0.7179487179
8	0.002312	0.006084	27.53692234	0.8717948718

10	0.003698	0.006561	36.04639828	1.061728395
12	0.005	0.006084	45.11006857	1.282051282
14	0.001568	0.012544	11.11111111	0.5
16	0.002312	0.017161	11.87284959	0.5190839695

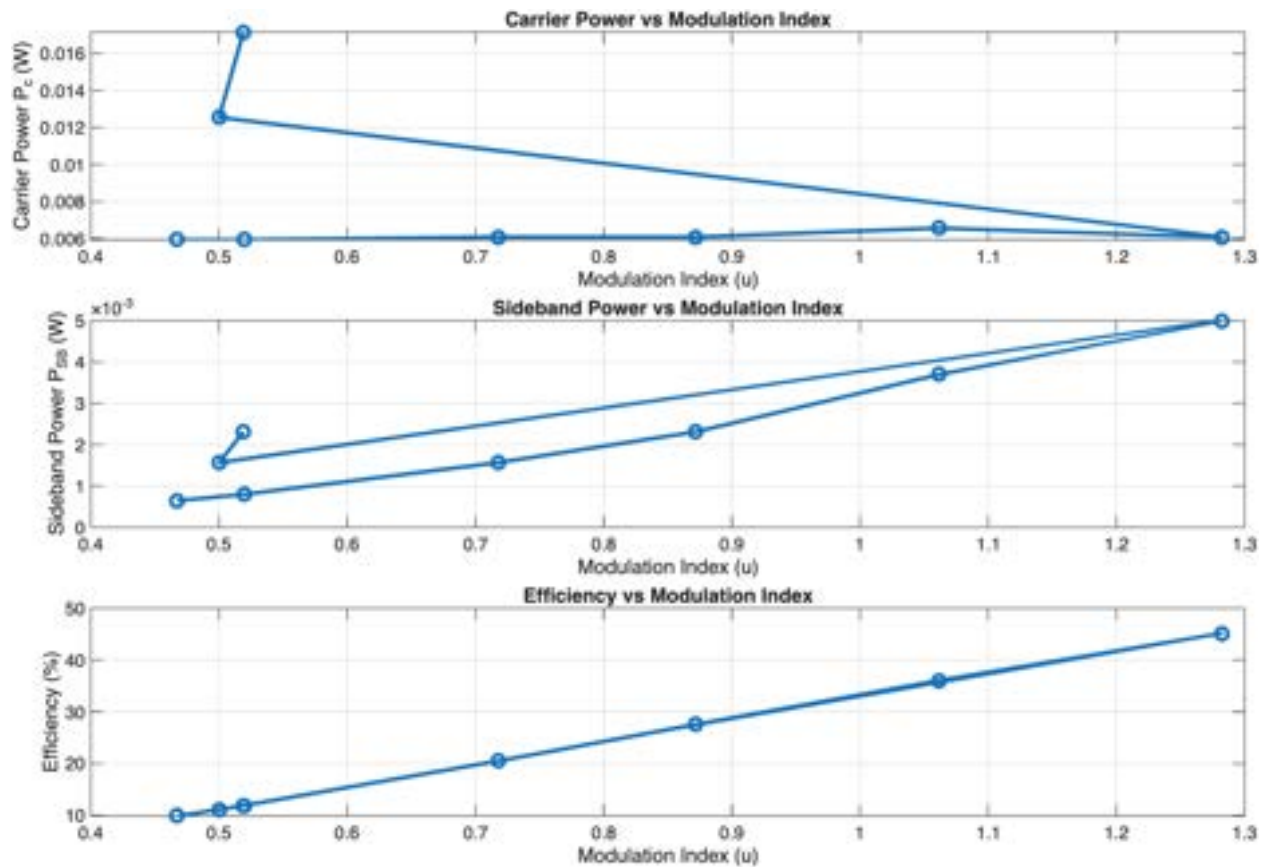


Figure B3.1 - P_{SB} , P_C , η Vs. μ

The plots show that the sideband power P_{SB} and efficiency η , increase as the modulation index increases. This is expected because P_{SB} it is also represented as $P_{SB} = \frac{\mu^2}{2} P_C$, which implies that P_{SB} is proportional to μ , thus the plot shows a gradually increasing slope.

The efficiency can be represented as $\eta = \frac{\mu^2}{2+\mu^2}$, which confirms that theoretically, the efficiency increases as the modulation index increases.

P_c can be represented as $\frac{A_c^2}{2}$, which does not contain a variable for the modulation index, thus P_c is independent of μ . Therefore, it is expected that the plot is horizontal because P_c is a constant. In Figure B3.1, we notice that P_c is mainly horizontal, which proves that P_c does not depend on the value of μ .

For all of the plots, we observe that at around $V_{pp} = 10$ V, the modulation index becomes distorted due to overmodulation. When the envelope becomes distorted, accurate measurements on the oscilloscope become more difficult, which is why the modulation index suddenly decreases to 0.5.

Problem B.4

For envelope detector to work, $A + m(t) \geq 0$
to ensure no overmodulation

$$[m(t) = A_m \cos(2\pi f_m t)]$$

$\hookrightarrow A_{\min} = -A_m$ ① 0.1
 $A - A_m \geq 0$
 $[A_m \leq 0.1]$

② if $\eta = \frac{\mu^2}{2 + \mu^2}$, then $\eta_{\max} = \frac{1}{2 + 1}$, when $\mu = 1$
the largest value of μ before overmodulation.

since $\mu = \frac{A_m}{A_c} = \frac{A_m}{0.1} \rightarrow A_m = (1)(0.1) = 0.1$

$$\therefore A_m = 0.1$$

$$[e_{AM}(t) = [0.1 + 0.1 \cos(2\pi f_m t)] \cos(2\pi f_c t)]$$

PART C - Demodulation of AM Signals

Problem C.1

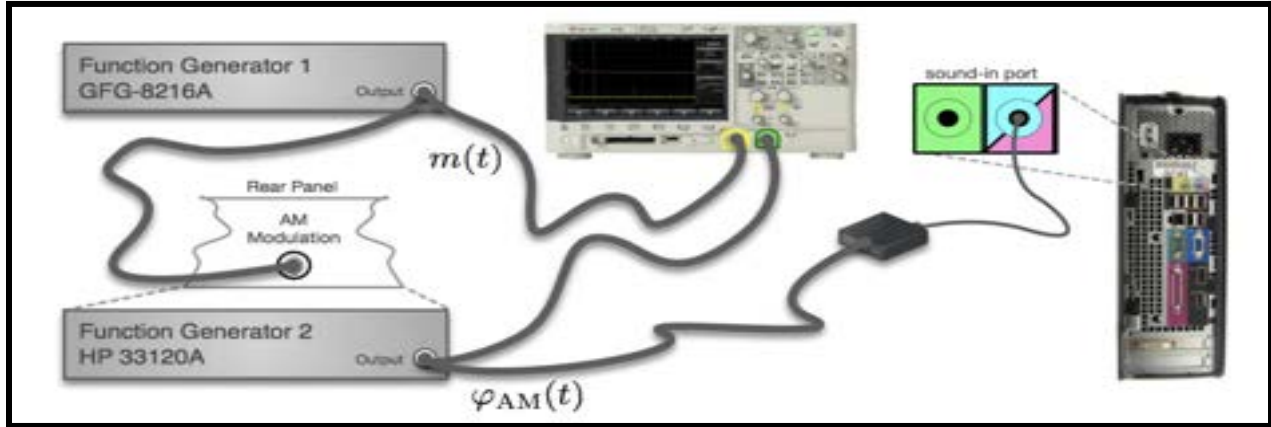


Figure C1.1: Connection Diagram for Part C

The modulation index μ describes how much the carrier amplitude varies relative to its unmodulated level A_c . Figure C1-B depicts the positive/negative swings of the amplitude and envelope functions for an AM signal.

Modulation Index:	$\mu = \frac{A_{\max} - A_{\min}}{2A_c}$	(A.6a)
Positive Modulation Index:	$\mu_+ = \frac{A_{\max} - A_c}{A_c}$	(A.6b)
Negative Modulation Index:	$\mu_- = \frac{A_c - A_{\min}}{A_c}$	(A.6c)

Figure C1.2: Modulation Index Equations derived in the Appendix



Figure C1.3: Function Generator & Function/Arbitrary Waveform Generator

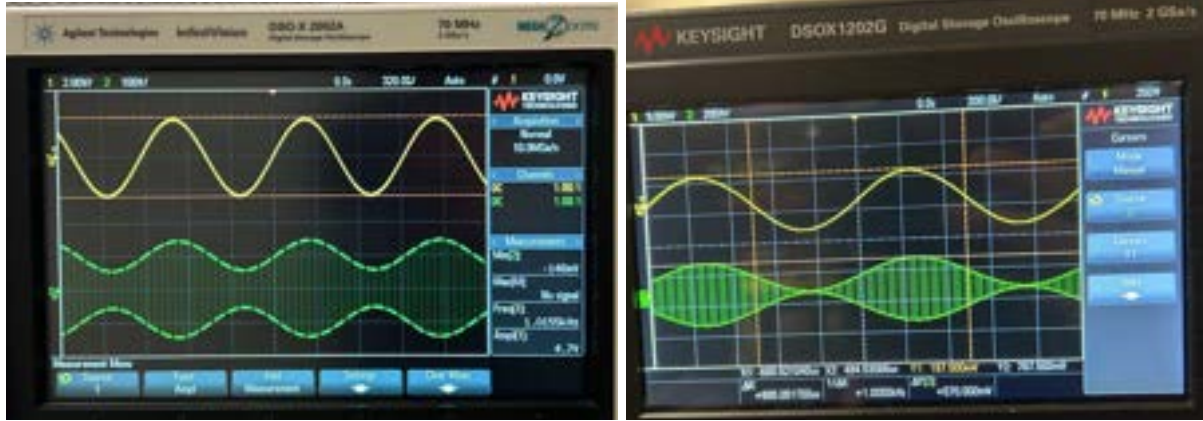


Figure C1.4: $\varphi_{AM}(t) = [A_c + Km(t)]\cos(2\pi f_c t) = [A_c + A_m \cos(2\pi f_m t)]\cos(2\pi f_c t)$

$$\mu = \frac{A_m}{A_c}$$

Equation C1.1: Modulation Index (μ) Ratio

- **Positive Modulation (μ_+):** Maximum upward swings of the signal envelope (Peaks)
- **Negative Modulation (μ_-):** Maximum downward swings of the envelope (Troughs)

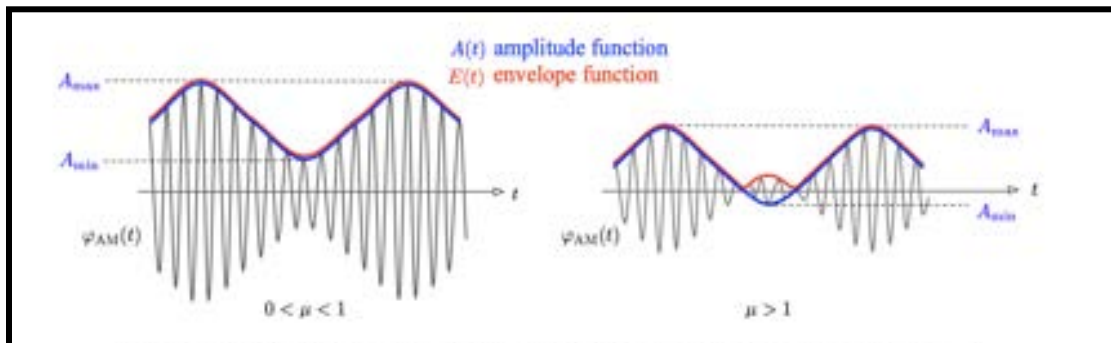


Figure C1.5: Amplitude and Envelope Functions derived from the Appendix

In the understanding of Asymmetric Modulation (AM), for AM broadcasting, the modulation index μ represents the ratio of the peak amplitude of the message signal to the amplitude of the unmodulated carrier. Figure C1-D displays the AM signal as described in step A1. In this case, based on the signal generated in Step A1 of this experiment, when $\mu_+ = 125\%$ the carrier's peak amplitude reaches $(2.25)A_c$. The negative modulation μ_- refers to the maximum downward swing toward the zero axis, as the troughs.

Therefore, it is possible to demodulate an AM radio signal with a positive modulation index $\mu_+ = 125\%$ using an envelope detector, provided that the negative modulation index μ_- remains $\leq 100\%$.

- **The Negative Limit (μ_-)**

This is the critical value for an envelope detector. An envelope detector works by tracing the peaks of the carrier wave. If μ_- exceeds 100%, the carrier amplitude actually hits zero and tries to go negative, causing a phase reversal. The envelope detector cannot track phase; it only sees the absolute value. This results in severe clipping and flat-topping at the zero-crossing point, creating massive audio distortion. Since the standard mandates $\mu_- \leq 100\%$, the carrier never crosses the zero line, and the shape of the message remains intact on one side of the axis.

- **The Positive Limit (μ_+)**

The standard allows μ_+ to reach 125%. This simply means the carrier is allowed to swing much higher than its resting state, reaching $(2.25)A_c$. As long as the electronic components in the receiver (diodes and capacitors) can handle the increased voltage without saturating or breaking down. The envelope detector will follow the upward curve. There is no phase reversal or distortion of the signal at the positive peak.

Parameter	Limit	Effect on Envelope Detector
Negative Modulation (μ_-)	100%	• Prevents phase reversal • Prevents pinching ○ For the signal at the zero axis.
Positive Modulation (μ_+)	125%	• The detector follows the higher peak • No distortion occurs ○ As long as the hardware isn't overdriven.

Table C1.1: Comparison of Maximum Modulation Indices (μ_+ & μ_-)

Therefore, as long as the carrier is not pinched off due to a negative modulation value of $\geq 100\%$, and can still have positive peaks ($\leq +125\%$) without losing the original modulating signal's integrity, the envelope detector will continue to be a useful (and inexpensive) method of demodulation for the carrier signal.

Phase inversion (or phase reversal) is the greatest failure point of a standard envelope detector (same as its rectifier type) because it is dependent on the modulation index crossing below zero to achieve phase reversal. Therefore, in the case of envelope detection, the modulation index can be greater than or equal to $\mu_+ = 125\%$ and still allow for proper function.

In public broadcasting, the technical standards allow for a maximum positive variation of 125% of the average level of a signal. This is an excellent way to increase the signal-to-noise ratio (SNR) and the loudness of a program without causing catastrophic distortion from excessive negative modulation. This technique is often achieved using asymmetric processing in the studio, in which, during a recording, the engineer assures that the peaks of the voice (which are naturally asymmetric) push the carrier more than they pull the carrier.

Envelope detectors will function properly as long as the total envelope $[A_c = m(t)]$ never becomes negative. Therefore, the standard limits negative modulation ($m(t) \leq \mu_- = 100\%$) such that the envelope of the signal will reach a precise value of zero; it will not cross through the zero value. Therefore, with respect to this envelope, the waveform does not fold or clip (which normally causes the high harmonic distortion experienced when a signal is over-modulated).

Problem C.2

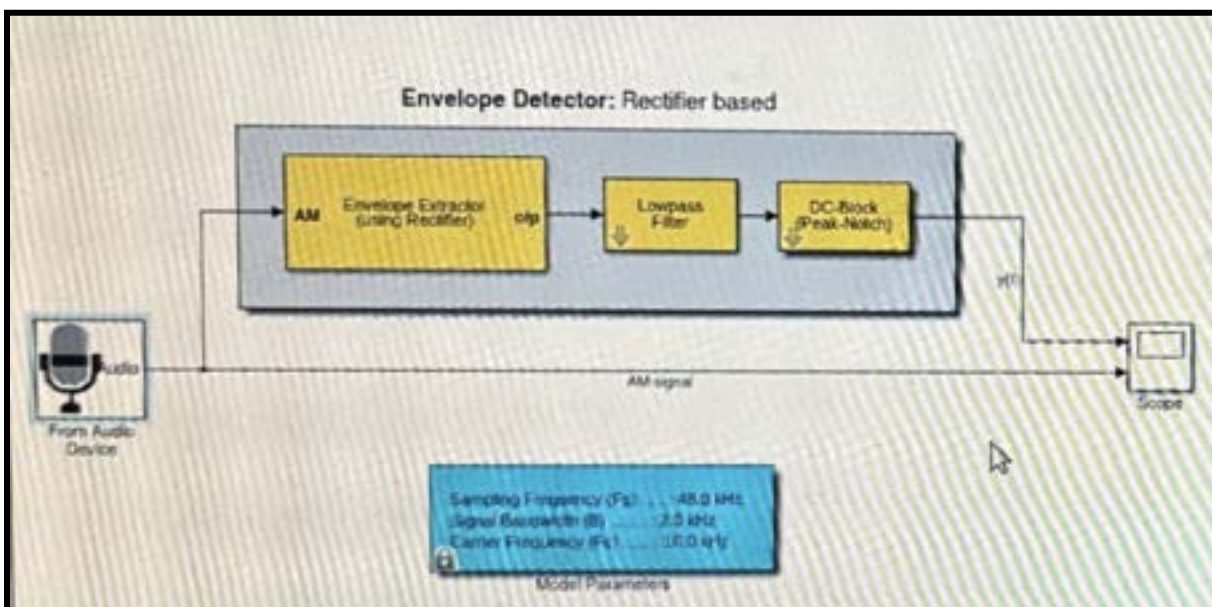


Figure C2.1: Rectifier-Based (*Exp2_EnvelopeDetectionRectifier.slx*)

- **Rectifier Detector:** This simple system will function without distortion because the signal never undergoes the phase reversals that lead to rectification errors.

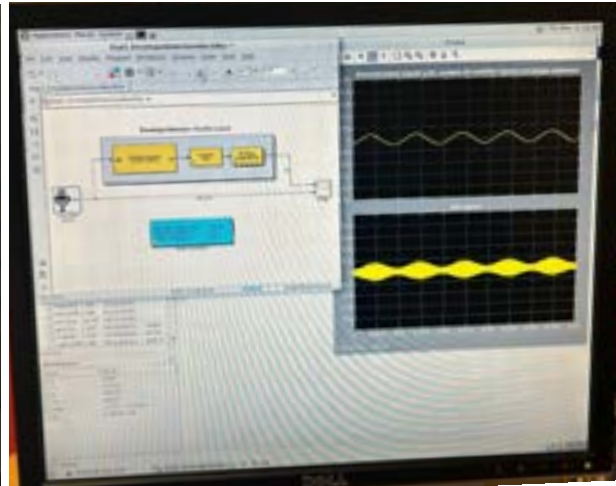
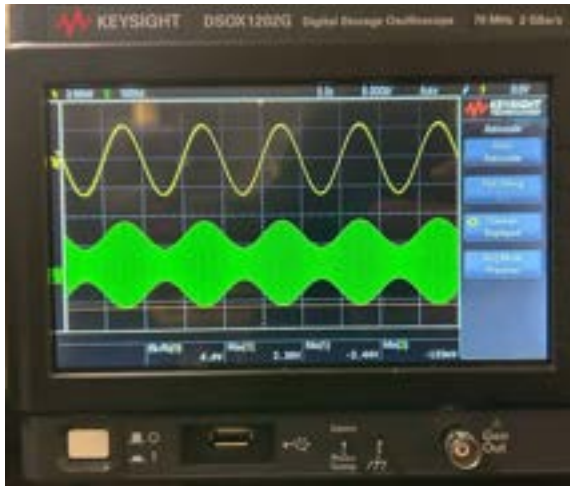


Figure C2.2: $0 \leq \mu \leq 0.25$

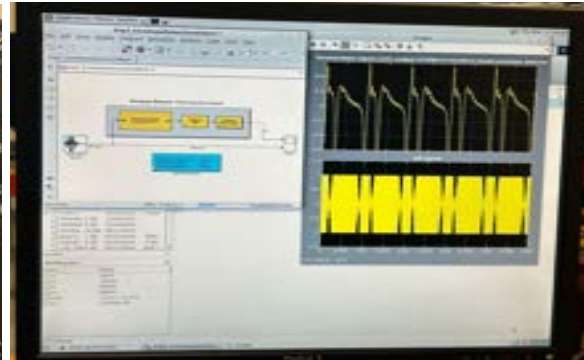
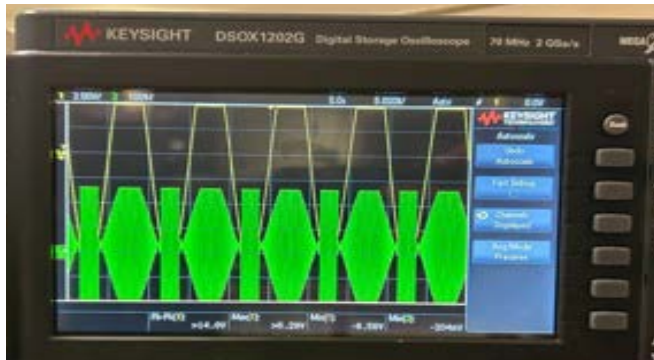


Figure C2.6: $\mu \geq 1.0$

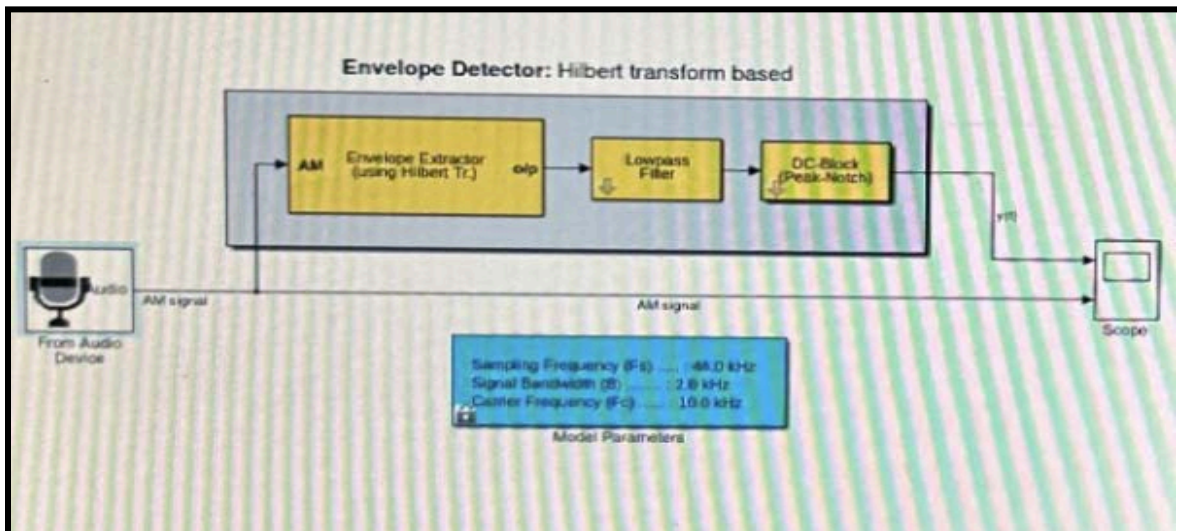


Figure C2.7: Hilbert-Transform Based (*Exp2_EnvelopeDetectionHilbert.slx*)

- Hilbert-based Detector:** As shown in the derivation, this system extracts $E(t) = |A_c + m(t)|$. Since $\mu_- = 100\%$ ensures $A_c + m(t) \geq 0$, the absolute value operator is effectively transparent, and the message $m(t)$ is recovered perfectly after the DC block.

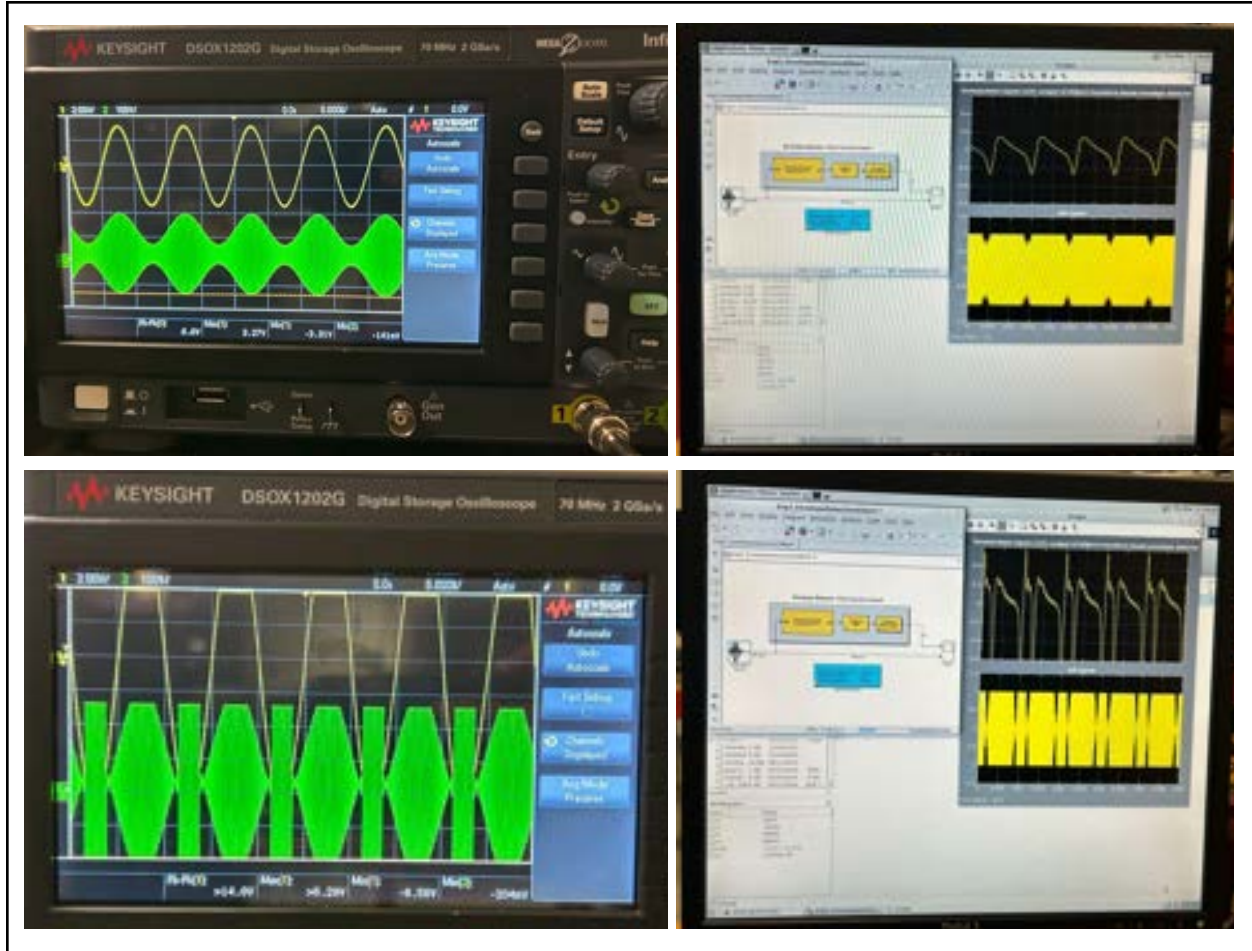


Figure C2.8: Demodulated Signal for Cases $\mu \leq 1$ and $\mu > 1$

For installation in inexpensive radio receivers, the recommended use is the Rectifier-based Envelope Detector. While both structures aim to extract the message signal from the carrier, the rectifier-based approach is the industry standard for low-cost hardware for several practical reasons:

- Circuit Simplicity & Cost**

In a physical circuit (Rectifier-based), this can be implemented with just a handful of passive components: a single diode, a resistor, and a capacitor. This makes it extremely cheap to manufacture and integrate into basic consumer electronics. As seen in the Simulink model (Hilbert Transform-based), this requires an Envelope Extractor block that performs complex

mathematical operations to create an analytic signal. In the analog domain, this is difficult and expensive to build; in the digital domain, it requires significant processing power (DSP) that would drive up the cost of an inexpensive receiver.

- **Implementation Efficiency**

The rectifier method simply slices the signal to remove the negative half-cycle and uses a lowpass filter to smooth the result. The Hilbert transform requires shifting the phase of every frequency component of the signal by 90° ($-\frac{\pi}{2}$ radians). Achieving this across the entire audio bandwidth requires complex wideband phase-shifting networks, which are not cost-effective for simple AM radios. Therefore, further implying that the rectifier-based envelope detector is suitable for installation in inexpensive radio receivers designed to receive AM broadcast radio signals.

Furthermore, it should heavily be noted that additional demodulated signal cases for a rectifier-based envelope detector are provided in the Appendix.

- **Performance vs. Requirement (Reliability)**

AM broadcast standards (as discussed earlier) ensure that the negative modulation index μ_- does not exceed 100%. Since the signal is guaranteed not to have phase reversals, the simple rectifier/diode detector performs more than adequately for clear audio reception. The added precision of a Hilbert-based detector provides no meaningful benefit to a listener using a budget speaker.

Feature	Rectifier-Based	Hilbert-Transform-Based
Component Count	Very Low (Diode + RC)	High (Complex logic/DSP)
Cost	Negligible	Moderate to High
Common Use	Pocket radios, car tuners	High-end SDRs, Lab equipment

Table C2.1: Demodulated Signal for Cases $\mu \leq 1$ and $\mu > 1$

Problem C.3

This Simulink model uses a large buffer to store incoming signals. Thus, resulting in high latency such that the effects of changes in the AM signal appear on the Simulink virtual scope only after a relatively long delay. Therefore, when making changes to the AM signal, as an advisory precautionary procedure, stop and restart the Simulink model to observe changes in detector outputs.

The Simulink model below in Figure C3-A (Exp2 EnvelopeAndCoherent.slx) is used to demodulate the AM signal; the model uses a coherent detector and an envelope detector. The coherent detector extracts the carrier $\cos(\omega_c t)$ from the AM signal using a Phase-Locked-Loop (PLL) set to the carrier frequency f_c . We specifically want to see how well the two detectors work when demodulating AM signals with various modulation indices.

To create an AM signal with a modulation index $\mu \approx 0.5$, set the modulating signal's amplitude $m(t)$ to $5 V_{pp}$. Launch the Simulink model. Record and plot the signals shown on the Simulink virtual scope after observing the demodulated signals at detector outputs. To produce an AM signal with a modulation index $\mu \approx 1.5$, the amplitude of the modulating signal $m(t)$ is set to $15 V_{pp}$. The Simulink model should be restarted. The signals shown on the Simulink virtual scope after observing the demodulated signals at detector outputs are then plotted.

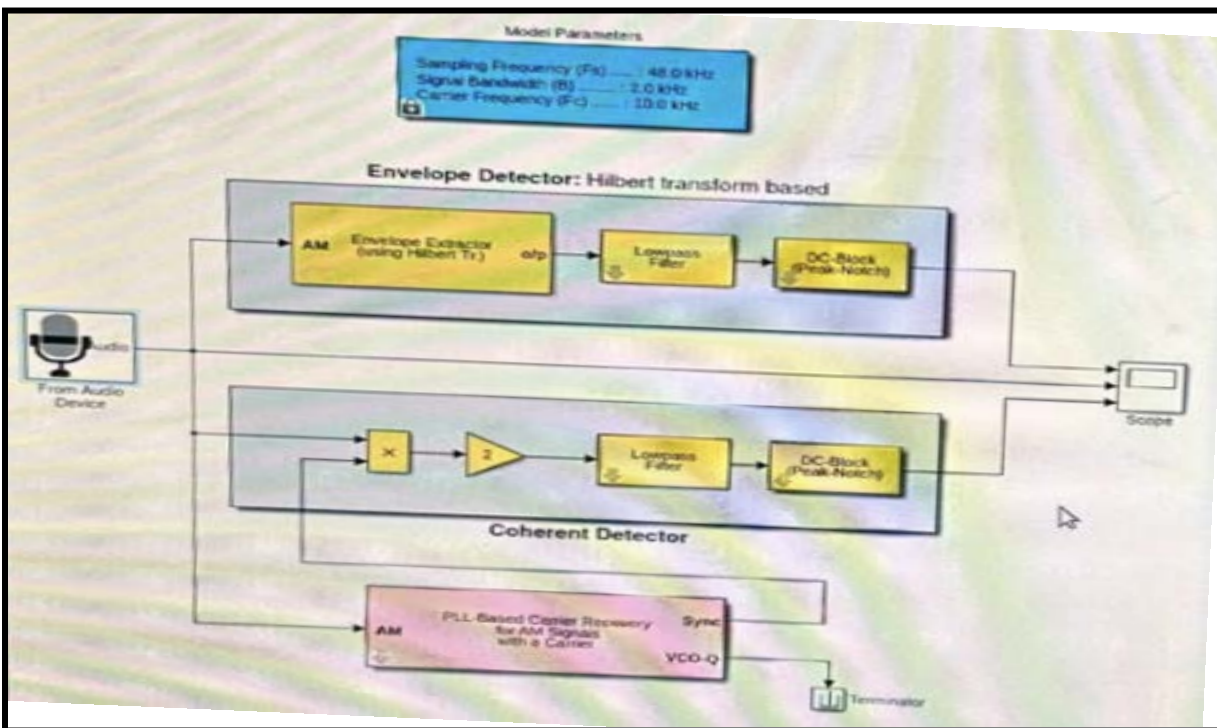


Figure C3.1: Exp2_EnvelopeAndCoherent.slx

In Step C.5, the core observation centers on how each detector handles overmodulation ($\mu > 1$). While both perform similarly under normal conditions, their underlying physics cause them to behave very differently once the signal breaks the 100% threshold.

- **Under-Modulation ($\mu \approx 0.5$)**

At this index, the message signal amplitude is half that of the carrier. Both detectors are successful. The outputs on the scope should look nearly identical (ignoring minor phase shifts or gain differences).



Figure C3.2: Demodulating an AM Signal using a Phase-Locked-Loop (PLL) tuned to the f_c

- **Over-Modulation ($\mu \approx 1.5$)**

When $\mu > 1$, the carrier envelope tries to go negative, resulting in phase reversals (this is where the distinction becomes clear).

- **Envelope Detector:**

- The envelope detector (whether Hilbert or Rectifier-based) can only track the magnitude (absolute value) of the signal. When the signal crosses zero, the detector flips the negative portion upward.
- On the scope, rectification distortion or clipping occurs where the troughs of the sine wave are chopped and inverted. The audio would be harsh and heavily distorted, implying failure.
- Therefore, successfully recovers the message signal. Since the envelope never crosses the zero axis, the shape of the message is perfectly preserved in the peaks of the carrier.

- **Coherent Detector:**

- The coherent detector multiplies the incoming signal by a locally generated, phase-locked carrier ($\cos(\omega_c t)$), it is capable of undoing the phase reversals.
- The multiplication process correctly handles the negative swings of the message signal. Even though the signal is technically overmodulated, the coherent detector recovers the original sine wave shape without the pinched distortion seen in the envelope detector and therefore implies success.

- Therefore, successfully recovers the message signal. The PLL locks to the carrier frequency and phase, allowing for clean multiplication and filtering.

Modulation Index (μ)	Envelope Detector	Coherent Detector
$\mu = 0.5$	Successful	Successful
$\mu = 1.5$	Failed (Distorted)	Successful

Table C3.1: Demodulating AM Signals with Different Modulation Indices

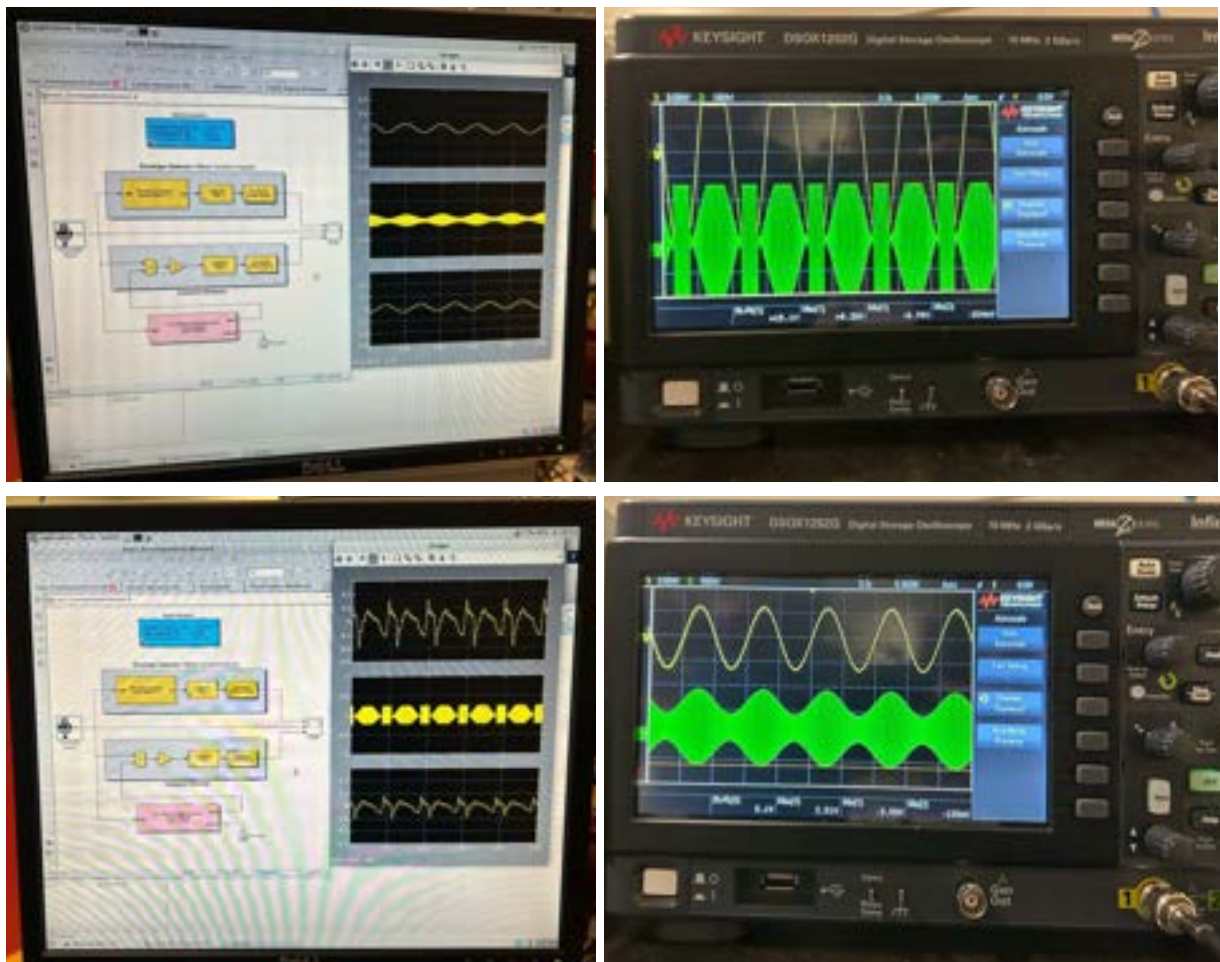


Figure C3.3: Envelope Detector and a Coherent Detector to Demodulate an AM Signal

The Envelope Detector is only successful when $\mu \leq 1$. It is a non-coherent receiver that relies on the physical shape of the carrier's peak.

The Coherent Detector is much more robust. It is successful regardless of the modulation index (provided the PLL can maintain a lock) because it uses the phase information of the carrier to correctly interpret the message signal, even during overmodulation.

Problem C.4

Replacing the Phase-Locked-Loop (PLL)-based carrier recovery with a simple local oscillator (LO) changes the receiver from a synchronous (locked) coherent detector to an asynchronous one. In this scenario, the output of the detector depends entirely on the phase and frequency alignment between the incoming carrier and your local oscillator.

To determine the output, let the incoming AM signal be:

$$s(t) = [A_c + m(t)]\cos(2\pi f_c t + \phi)$$

- Where ϕ is the phase of the received carrier.

The local oscillator generates:

$$c(t) = \cos(2\pi f_c t + \theta)$$

When multiplying these together (as shown in the Coherent Detector block of the Simulink model), the product is:

$$y(t) = [A_c + m(t)]\cos(2\pi f_c t + \phi)\cos(2\pi f_c t + \theta)$$

Using the product-to-sum trigonometric identity, the signal before the Lowpass Filter is:

$$y(t) = \frac{1}{2} [A_c + m(t)][\cos(\phi - \theta) + \cos(4\pi f_c t + \phi + \theta)]$$

After the Lowpass Filter removes the high-frequency $2f_c$ component and the DC-Block removes the A_c constant, the final output is:

$$\text{Output} \approx \frac{1}{2} m(t)\cos(\Delta\phi)$$

- Where $\Delta\phi = \phi - \theta$ is the phase error between the carrier and the local oscillator.

Theoretically, it can be used to demodulate the signal, but practically, no, unless it can ensure perfect synchronization. There are two cases of discrepancies:

1. Phase Error ($\Delta\phi$)

- The output amplitude is scaled by $\cos(\Delta\phi)$.
- If the phase difference is 90° ($\frac{\pi}{2}$ radians), the output becomes zero.
- This is known as null effect distortion, where the message signal completely vanishes even though the station is broadcasting.

2. Frequency Drift (Δf)

- In a real-world scenario, the local oscillator and the transmitter will have slight differences in frequency.
- If there is a small frequency offset Δf , the phase error becomes time-varying: $\Delta\phi = 2\pi\Delta f t$.
- The resulting audio will beat or pulsate in volume as the $\cos$ term oscillates between 1 and -1, making the audio unlistenable.

While a local oscillator can technically produce the message signal (thus, demodulating the AM signal), it lacks the Phase Locked Loop (PLL) seen in this model. The PLL is vital because it constantly adjusts the local oscillator to track the incoming carrier's phase ($\Delta\phi = 0$), ensuring $\cos(\Delta\phi) = 1$ and maintaining a steady, maximum-strength output. Without it, the signal will likely suffer from fading or total silence.

Conclusions

The examination of rectangular pulses and periodic signals in the frequency domain verified that spectral breadth and harmonic distribution are directly determined by time-domain limitations, particularly by the inverse relationship between pulse width and bandwidth. By evaluating the modulation index (μ), it was shown that frequency-domain analysis utilizing the one-sided RMS spectrum (Method 3) offers better precision for low modulation levels, although time-domain measurements (V_{max}/V_{min}) offer a fast assessment. Additionally, the data confirmed that although sideband power and transmission efficiency rise as μ approaches 1.0, surpassing this barrier causes carrier phase reversal and substantial harmonic distortion, making conventional time-domain measurements inaccurate.

The study identified a crucial trade-off between receiver cost and performance robustness with regard to demodulation. For standard broadcasting, where negative modulation is rigorously capped at $\mu_- = 100\%$, the rectifier-based envelope detection, proved to be an effective and economical option. Because the envelope never crosses the zero-axis, low-cost diode-based circuitry can be used even when positive peaks exceed $\mu_+ = 125\%$. The coherent detector, on

the other hand, proved to be superior in high-stress situations. It recovered the message signal even under overmodulation ($\mu = 1.5$), where envelope detectors failed because of rectification errors, by using a Phase-Locked Loop (PLL) to maintain carrier synchronization. In the end, the experiment demonstrated that although commercial receivers can identify envelope applications that need immunity to overmodulation distortion, coherent detection must be used for phase integrity.

Therefore, the experimental results imply that the basic features of Amplitude Modulation (AM) and the real-world constraints of different demodulation methods were effectively illustrated by the results from Parts A, B, and C.

References

Lathi, B. P., and Zhi Ding. *Modern Digital and Analog Communication*. Oxford University Press, Incorporated, 2021. Accessed 18 March 2026.

Zhao L., *ELE635: Communication Systems*. Toronto Metropolitan University, W2026. Accessed 11 March 2026.

Zhao L., *Exp3 Frequency Modulation and Demodulation*. Toronto Metropolitan University, W2026. Accessed 11 March 2026.

Appendix

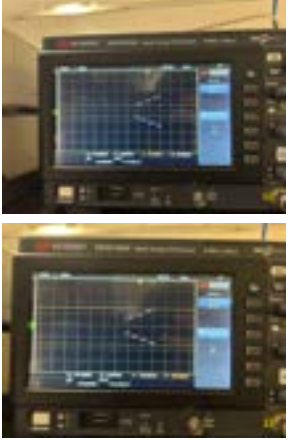
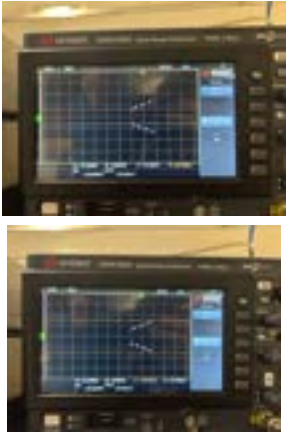
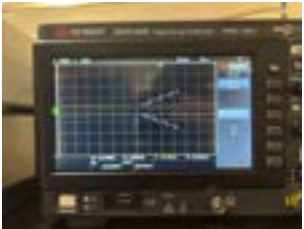
Below are the tabulated results for the experimental procedure of Part B (Modulation Index), covering all three (3) methods of analysis:

- **Method 1**

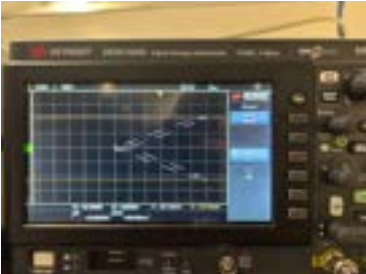
Amplitude V_{PP}	A_{max}	A_{min}
4	2.04	-145 mV
5	2.4	-155 mV
6	3.10	-167mV
8	3.9	-185 mV
10	4.9	-210 mV

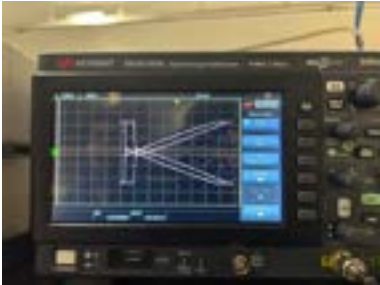
12	6.2	-231 mV
14	7.4	-27.5 mV
16	8.0	-30.0 mV

• **Method 2**

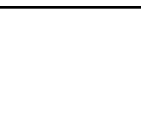
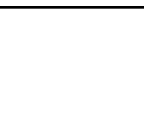
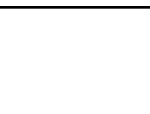
Amplitude V_{pp}	α Alpha (left side)	β (Right side)	Images
5	110 mV	310 mV	
4	125 mV	300 mV	
6	80 mV	330 mV	

			
8	35 mV	370 mV	 
10	5.00 mV	425 mV	 

12	35 mV	450 mV	 
14	50 mV	480 mV	 
16	50 mV	470 mV	

			
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• Method 3

Amplitude V_{PP}	Signal - S (V_{rms})	Carrier - C (V_{rms})	Signal - S (dB)	Carrier - C (dB)	F_s (Hz)	F_r (Hz)
5	20 mV 	77 mV 			8.6 Hz	10.9
4	18 mV 	77 mV 	12.685 	23.907 	8.86 Hz	10.9
6	28 mV 	78 mV 	23.5 	22.7 	8.9 Hz	10.9
8	34 mV 	78 mV 	15.625 	23.125 	8.5 Hz	10.8
10	43 mV 	81 mV 	17.5 	21.8 	8.8 Hz	10.5
12	50 mV 	78 mV 	18.7 	23 	8.76 Hz	10.9
14	28 mV 	112 mV 	15 	26.7 	8.96 Hz	11.0

						
16	34 mV 	131 mV 	18.7 	30 	8.9 Hz	10.9

- Problem C2 (Rectifier-Based)

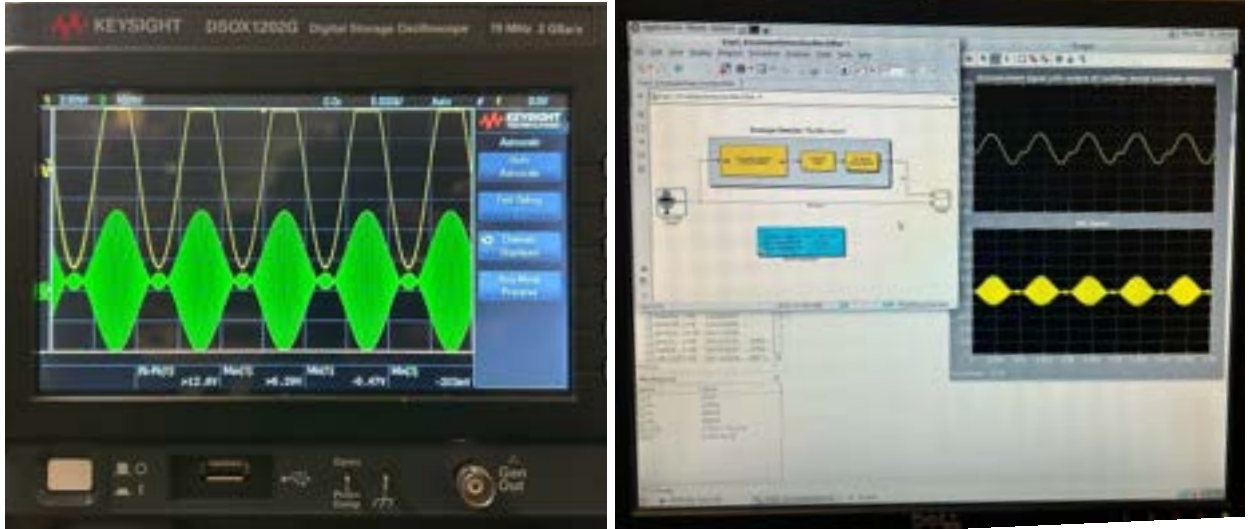


Figure C2.3: $0.25 \leq \mu \leq 0.50$



Figure C2.4: $0.50 \leq \mu \leq 0.75$

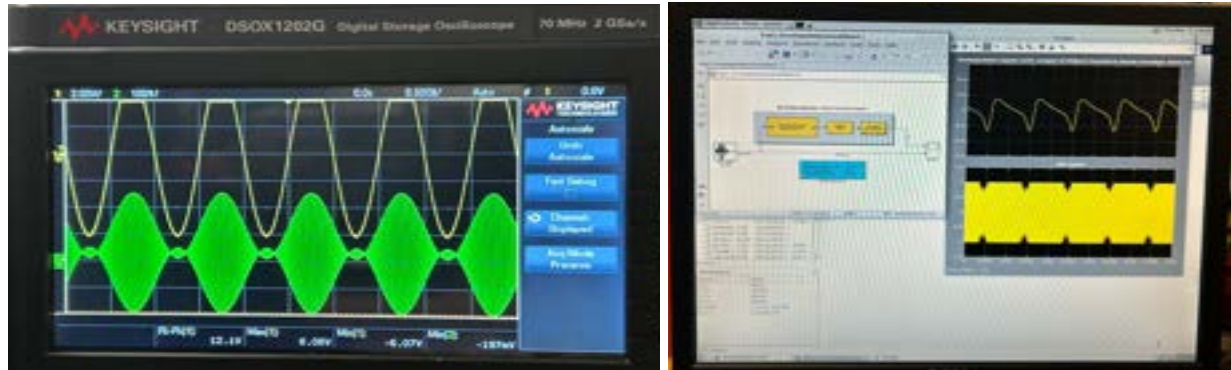


Figure C2.5: $0.75 \leq \mu \leq 1.0$